

Stuti Shah

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PROFESSIONAL EXPERIENCE

Data Science Intern, Volvo Group | Greensboro, NC May 2025 - Aug 2025

- Automated DMO file generation using Streamlit, reducing processing time from 4 hours to under 6 minutes and delivering \$176K in annual cost savings across distributed engineering teams.
- Designed an end-to-end ML analytics platform using Python, PostgreSQL, and Flask, linking CVM signals to millions of sensor records to enable scalable predictive performance analysis.
- Built and validated a Random Forest model to predict truck component damage and acceleration, supporting data-driven test readiness decisions across multiple vehicle platforms and configurations.
- Integrated a GHG emissions prediction pipeline into Power BI, estimating missing CdA values and enabling scenario-based emissions classification for regulatory compliance and sustainability reporting.

AI Intern, AquaOrange | Bangkok, Thailand Jun 2024 - Aug 2024

- Implemented AI testing workflows with BrowserStack and testRigor for 10+ Zoho CRM customizations, reducing QA cycles and speeding up releases.
- Automated regression testing pipelines, enhancing defect detection in critical CRM workflows and improving system reliability during feature iterations.
- Collaborated with product and engineering teams to convert testing insights into actionable improvements, enhancing data-driven decision-making for software upgrades.

Data Science Intern, HERE! Wireless USA | Dallas, TX Jun 2023 - Jul 2024

- Executed 50+ alpha testing workflows for Android and iOS, enhancing feature stability and reducing post-release issues.
- Analyzed feedback from 200+ users, providing insights for feature prioritization and UX improvements.
- Developed four pricing models to support leadership in minimizing revenue leakage and optimizing market positioning.

EDUCATION

Virginia Tech, Blacksburg, VA - Master of Engineering, Computer Science May 2026

Virginia Tech, Blacksburg, VA - Bachelor's, Computer Science May 2025

PROJECTS

HybridDiff AR: Hybrid Diffusion-Autoregressive Model for Efficient and High-Quality Image Generation

- Developed a hybrid image generation pipeline, merging a 33M parameter DDPM with transformer refinement for 250× faster inference on CelebA 64×64.

Multimodal Uncertainty-Aware Stock Return Forecasting After Earnings Calls

- Developing a multimodal model for next-day post-earnings stock returns, integrating transcripts, financial indicators, and social sentiment with cross-attention, and assessing uncertainty through ECE and conformal prediction intervals.

YOLOv8 Shelf Detection for Planogram Compliance (Items + Empty Spaces)

- Developing a shelf monitoring pipeline with YOLOv8 for product detection and empty shelf identification, utilizing public datasets for planogram compliance and restocking insights.

SKILLS

Programming & Scripting : Python

Data Science & Machine Learning : Model Evaluation, Statistical Analysis, Experiment Design, ML Fundamentals

AI / Deep Learning Frameworks : PyTorch, scikit-learn, DDIM Sampling, Transformer Architectures

Data Engineering & Databases : Data Modeling, ETL Pipelines, MySQL, PostgreSQL, Large-Scale Data Processing

Analytics & Visualization : Power BI, Tableau, Streamlit, Information Visualization, Dashboard Design

MLOps & Cloud-Native Tools : Docker, Kubernetes, Argo CD, CI/CD Pipelines, Model Deployment & Monitoring

Software Engineering Practices : Object-Oriented Design, Unit Testing, Git/GitHub, Linux